Multi-Player Games and Nash Equilibrium

Strategic Interactions in Game Theory

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Foundations for Economics, AI, and Strategic Decision-Making

Why Study Multi-Player Games?

Real-World Applications

Most strategic situations involve **more than two players**:

- **Markets:** Many firms competing
- **Traffic networks:** Many drivers choosing routes
- **Cybersecurity:** Multiple attackers and defenders
- **Multi-agent AI:** Autonomous systems interacting

Central Question

How should a rational player act when outcomes depend on **everyone else's** decisions?

Game theory provides the mathematical framework to analyze these interactions.

What Is a Strategic (Normal-Form) Game?

Definition

A **strategic game** is defined by three components:

1. **Players** (N)
2. **Strategies** (S_i for each player i)
3. **Payoffs** (u_i for each player i)

Mathematical Representation

We represent the game as:

$$\boxed{G = (N, (S_i)_{i \in N}, (u_i)_{i \in N})}$$

This compact notation fully describes the strategic environment.

Players (Set N)

Player Set

We assume there are n players, indexed as:

$$N = \{1, 2, \ldots, n\}$$

Assumptions:

- Each player is **rational**
- Players maximize their own payoff
- Common knowledge of the game structure

Examples:

- Firms in a market
- Countries in trade
- Autonomous agents
- Network nodes

Strategy Sets

Individual Strategy Set

Each player i has a **finite set of strategies**:

$$S_i = \{s_i^1, s_i^2, \ldots, s_i^{k_i}\}$$

where k_i is the number of available strategies for player i.

Joint Strategy Space

The **full strategy space** is the Cartesian product:

$$\boxed{S = S_1 \times S_2 \times \cdots \times S_n}$$

This contains **all possible outcomes** of the game.

If each player has 2 strategies and $n = 3$, then $|S| = 2^3 = 8$ possible outcomes.

Strategy Profile

Definition

A **strategy profile** specifies one strategy for each player:

$$s = (s_1, s_2, \ldots, s_n) \in S$$

where $s_i \in S_i$ is the strategy chosen by player i.

Interpretation

- Each profile s represents one **complete outcome**
- Every player has made a decision
- The profile determines the payoff for each player

Example

Payoff Functions

Definition

Each player i has a **payoff function**:

$$u_i : S \to \mathbb{R}$$

This maps each strategy profile to a real number representing player i's utility.

Interpretation

- $u_i(s)$ tells us how good outcome s is for player i
- Higher values are better
- May represent: money, performance, satisfaction, utility
- Different players may have different preferences over the same outcomes

Separating One Player from the Others

Alternative Notation

We rewrite the strategy profile as:

$$s = (s_i, s_{-i})$$

where:

- s_i is player i's strategy
- $s_{-i} = (s_1, \ldots, s_{i-1}, s_{i+1}, \ldots, s_n)$ is the strategy profile of **all other players**

Why This Matters

This notation allows us to analyze:

- What happens when **one player changes** their strategy
- While everyone else stays the same
- Essential for defining Nash equilibrium

Unilateral Deviation

Definition

A **unilateral deviation** occurs when only **one player** changes their strategy:

$$(s_i^*, s_{-i}^*) \quad \rightarrow \quad (s_i', s_{-i}^*)$$

where $s_i' \neq s_i^*$ and all other players keep their strategies fixed.

Key Question for Nash Equilibrium

Is this deviation profitable?

- If $u_i(s_i', s_{-i}^*) > u_i(s_i^*, s_{-i}^*)$: Player i benefits from deviating
- If $u_i(s_i', s_{-i}^*) \leq u_i(s_i^*, s_{-i}^*)$ for all s_i': Player i has no incentive to deviate

Nash equilibrium is based entirely on analyzing such deviations.

Best Response (Definition)

Definition

The **best response** of player i to others' strategies s_{-i} is:

$$\boxed{BR_i(s_{-i}) = \arg\max_{s_i \in S_i} u_i(s_i, s_{-i})}$$

Interpretation

- Given what everyone else does (s_{-i})
- What is the best possible action I can take?
- May be a **set** if there are multiple equally good responses

Example

If others play s_{-i}, and player i has strategies $\{A, B, C\}$:

Nash Equilibrium (Formal Definition)

Definition (Nash Equilibrium)

A strategy profile $s^* = (s_1^*, s_2^*, \ldots, s_n^*)$ is a **Nash Equilibrium** if:

$$\boxed{u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*) \quad \forall i \in N, \forall s_i \in S_i}$$

Interpretation

- No player can **improve** by unilaterally deviating
- Each player is doing the best they can, given what others do
- The system is in **strategic balance**

Key Property

Stability: If players reach Nash equilibrium, no one has an incentive to change strategy.

Nash Equilibrium Using Best Responses

Theorem

A strategy profile s^* is a Nash Equilibrium if and only if:

$$\boxed{s_i^* \in BR_i(s_{-i}^*) \quad \forall i \in N}$$

Interpretation

- Every player is already playing a **best response** to the others
- No profitable unilateral deviation exists
- **Mutual best response** condition

Practical Use

To find Nash equilibrium:

1. Compute $BR_i(s_{-i})$ for each player i

Nash Equilibrium as a Fixed Point

Definition

Define the **global best-response operator**:

$$BR(s) = BR_1(s_{-1}) \times BR_2(s_{-2}) \times \cdots \times BR_n(s_{-n})$$

Theorem

s^* is a Nash Equilibrium if and only if:

$$\boxed{s^* \in BR(s^*)}$$

Mathematical Insight

- Nash equilibrium is a **fixed point** of the best-response mapping
- Links game theory to fixed-point theory

Example: 3-Player Coordination Game

Setup

- **Players:** $N = \{1, 2, 3\}$
- **Strategies:** Each player chooses location A or B
- **Payoffs:**
  - If **all 3** choose the same location: each gets **10**
  - If **2** coordinate and **1** is alone:
    - The pair gets **7** each
    - The lone player gets **0**

Example (Sample Outcomes)

- (A, A, A): All get 10
- (A, A, B): Players 1,2 get 7; Player 3 gets 0
- (A, B, A): Players 1,3 get 7; Player 2 gets 0

Checking Nash Equilibrium in the Example

Consider Profile: (A, A, A)

Current payoffs: $(10, 10, 10)$

Check each player's deviation incentive:

- **Player 1 deviates to B:** $(B, A, A) \rightarrow$ Player 1 gets $0 < 10$ (no improvement)
- **Player 2 deviates to B:** $(A, B, A) \rightarrow$ Player 2 gets $0 < 10$ (no improvement)
- **Player 3 deviates to B:** $(A, A, B) \rightarrow$ Player 3 gets $0 < 10$ (no improvement)

No player benefits from deviating!

Conclusion

$$\boxed{(A, A, A) \text{ is a Nash Equilibrium}}$$

By symmetry, (B, B, B) is also a Nash Equilibrium.

Key Properties of Nash Equilibrium

Strategic Stability

- Represents a **stable state** — no unilateral deviation is profitable
- Self-enforcing without external enforcement

Not Always Socially Optimal

- Nash equilibrium may be **inefficient**
- Example: Prisoner's Dilemma, Tragedy of the Commons
- Individual rationality $\neq$ collective optimality

Multiplicity and Mixed Strategies

- May be **unique** or **multiple** equilibria
- Sometimes only exists in **mixed strategies**
- Coordination problems arise from multiple equilibria

Existence of Nash Equilibrium

Theorem (Nash Existence Theorem, 1950)

Every finite n-player game has at least one Nash Equilibrium in **mixed strategies**.

Implications

- Strategic stability **always exists** mathematically
- May require randomization (mixed strategies)
- Pure strategy equilibrium is not guaranteed

Proof Technique

- Uses **Kakutani fixed-point theorem**
- Shows best-response correspondence has a fixed point
- Foundation of modern game theory

Summary: The Full Modeling Process

Steps to Solve Any Multi-Player Game

1. **Define players:** $N = \{1, 2, \ldots, n\}$
2. **Define strategies:** S_i for each player i
3. **Define payoffs:** $u_i : S \to \mathbb{R}$ for each player i
4. **Construct strategy profiles:** $s = (s_1, \ldots, s_n)$
5. **Compute best responses:** $BR_i(s_{-i})$ for each player
6. **Identify Nash equilibrium:**

$$\boxed{s^* \in BR(s^*)}$$

Core Insight

Nash equilibrium provides a **rigorous mathematical framework** for analyzing strategic interactions in multi-agent systems.

Thank You!

Questions?

Next: Mixed Strategies and Bayesian Games